Guesstimate Masterclass: Core Concepts

Welcome to the Guesstimate Masterclass! This guide covers the fundamental frameworks you need to break down any guesstimate problem: understanding the Demand vs. Supply constraints and building a logical structure using the MECE principle.

Demand vs. Supply Approach

Supply and demand are two key concepts in economics that can be used in guesstimates to estimate the potential outcomes of a situation. Both approaches involve making estimates based on available data, but they focus on different aspects of the problem.

The Supply-Side Approach Bottom UP

The **supply-side approach** focuses on the factors that affect the ability of a company or economy to produce and supply goods and services. This approach is often used to evaluate the potential impact of changes in production costs, technology, and the availability of inputs (like labor and raw materials) on the ability to produce and supply.

This approach is advantageous when the factors involved in production are **well-defined and measurable**.

→ Operating Capacity

- **Example:** Estimating the number of cars refueled at a single gas station. You would focus on supply constraints like:
  - Number of pumps
  - Time taken to refuel one car
  - Operating hours of the station

The Demand-Side Approach Top Down

The **demand-side approach**, on the other hand, focuses on the factors that affect the desire and ability of consumers to buy goods and services. This approach is used to evaluate the potential impact of changes in consumer income, preferences, and prices on the demand for those goods and services.

- **Example:** Estimating the number of Netflix subscriptions in a city. You would focus on demand factors like:
  - Number of households
  - Percentage with internet access
  - Percentage with sufficient disposable income
  - Percentage interested in streaming services

How to Use Both

It is essential to consider **both sides** in any guesstimate. One side often acts as the primary calculation, while the other serves as a crucial sanity check.

- **The Sanity Check:** The demand side provides information about the potential market and can help validate supply-side estimates.
- **Example:** A supply-side calculation might suggest a single coffee shop *could* serve 1,000 customers per day based on its capacity. However, if a demand-side check shows there are only 500 potential coffee drinkers in the immediate area,

the supply-side estimate is irrelevant. The *actual* number is constrained by demand.

A good rule of thumb is to start from the side that is easier to calculate (often the supply side, as it's more tangible) and then perform a sanity check using the other (demand) side.

MECE: Top-Down vs. Bottom-Up

The most important principle for structuring any guesstimate is **MECE** (pronounced "mee-see").

MECE stands for **Mutually Exclusive, Collectively Exhaustive**.

- **Mutually Exclusive:** Your categories have no overlaps. You don't double-count anything. (e.g., You can't be in both the "0-14" and "15-25" age groups).
- **Collectively Exhaustive:** Your categories cover all possibilities. You don't miss anything. (e.g., The age groups 0-14, 15-25, 26-59, and 60+ cover the entire population).

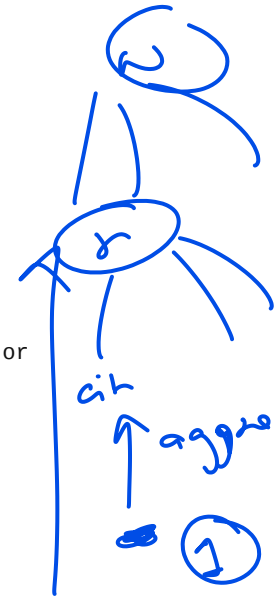
Applying the MECE principle prevents you from making structural errors. The two most common ways to build a MECE structure are the Top-Down and Bottom-Up approaches.

Top-Down Approach (The "Big Picture" Filter)

This approach starts with the largest possible group (like the total population) and uses a series of filters (percentages, assumptions) to narrow it down to the final answer. This is often linked to a **demand-side** analysis.

Example: *Estimate the number of tennis balls sold in India per year.*

1. Start with the **Population of India** (e.g., 1.4 billion).
2. Filter by **Age:** What % play sports? (e.g., 15-59 age group).
3. Filter by **Income:** What % can afford to play tennis? (e.g., Middle and Upper Income).
4. Filter by **Interest:** What % of that group *actually* plays tennis? (e.g., 1%).
5. Filter by **Frequency:** How many balls does an average player use per year? (e.g., 12 balls/year).
6. Calculate: (Population X % Age X % Income x % Interest X Balls/Player) = Total Balls



Bottom-Up Approach (The "Single Unit" Scale-Up)

This approach starts with a single, representative unit (like one shop, one person, or one gas pump) and scales it up to find the total. This is often linked to a **supply-side** analysis.

Example: *Estimate the revenue of a single movie theater (multiplex) in a month.*

1. Start with **One Screen**.
2. Calculate **Capacity:** How many seats per screen? (e.g., 150 seats).
3. Calculate **Showings:** How many shows per day? (e.g., 5 shows).
4. Calculate **Occupancy:** What is the average occupancy rate? (e.g., 50%).
5. Calculate **Ticket Price:** What is the average ticket price? (e.g., ₹250).
6. Calculate **Daily Revenue per Screen:** (Seats X ShowsX Occupancy % X Price)

7. Scale up to **Total Revenue:** (Daily Revenue per Screen X Number of Screens X 30 days)

Tying It All Together

The best guesstimates often use **both methods**.

- You might use a **Bottom-Up** (supply-side) approach to estimate the number of pizzas a single Domino's store can sell.
- Then, you'd use a **Top-Down** (demand-side) approach to check if the total pizza demand in that city makes sense.
- If your Bottom-Up estimate (total pizzas from all stores) is vastly larger than your Top-Down estimate (total people willing to buy pizza), you know one of your assumptions is wrong.

Commonly Used Assumptions for Guesstimates

Note: These are general figures. Always state your assumptions clearly.

Population

- **World:** 8 billion
- **India:** 1.4 billion (140 Cr)
- **USA:** 330 million (33 Cr)
- **UK:** 68 million (6.8 Cr)
- **Canada:** 40 million (4 Cr)
- **Delhi:** 30 million (3 Cr)
- **Bangalore:** 10 million (1 Cr)

Household Size

- **USA:** 2.5
- **India:** 4.4 (~ 4 in urban and ~ 5 in rural areas)

Demographic Splits (Global)

- **Rural-Urban Split:** 60% Rural / 40% Urban
- **Gender Split:** 50% Female / 50% Male
- **Occupation (Working Age):** 50% of total population
  - Of which: 75% Blue Collar / 25% White Collar

Demographic Splits (India)

- **Rural-Urban Split:** 70% Rural / 30% Urban
- **Age Split:**
  - 0-14: 25%
  - 15-25: 20%
  - 26-59: 40%
  - 60+: 15%
- **Income-wise Split (Urban):**
  - Below Poverty Line (BPL): 0%
  - Lower Income: 40%
  - Middle Income: 40%
  - Upper Income: 20%
- **Income-wise Split (Rural):**

- BPL: 20%
- Lower Income: 30%
- Middle Income: 40%
- Upper Income: 10%

- **Occupation (Working Age):** 50% of total population
  - *View 1:* 75% Blue Collar / 25% White Collar
  - *View 2:* 40% Agriculture / 25% Industry / 35% Services
  - *View 3:* 70% Unorganised Sector / 30% Organised Sector

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condom

How many cups of tea are consumed in all of India every week?

Setting the Stage

We'll assume that:

- Every day of the week is considered identical and does not affect tea consumption.
- The number of people who take tea more than twice a day is negligible.
- Consumption varies depending on the age group; hence we'll use an age-wise split to estimate the weekly consumption for the whole population.

Approach

The total population of India is 140 Cr. We've assumed that most people in the age group 0-14 would prefer milk or coffee, which is evenly distributed in the 15-25 group. Tea is typical in older age groups. Hence, we can find the average daily consumption per person as follows:

Age group	(A) % of people of that age group	(B) No of cups per day	(C) Further division	Daily Consumption (A x B x C)
0-14	**25%**	0	90%	0
		1	10%	0.025
15-25	**20%**	0	40%	0
		1	50%	0.1
		2	10%	0.04
25+	**55%**	0	10%	0
		1	50%	0.275
		2	40%	0.44

Summary Calculation	Value
Daily Consumption per person	**0.88**

Weekly Consumption in Cr	**862.4** (0.88 X 7 X 140)

Hence, ~ 850 Cr cups of tea are consumed in India weekly.